WEEK 7

Date: 10/09/2026

Title: Method Overriding using RBI and Different Banks

Objective / Aim

To design and implement a Java program to demonstrate **method overriding and runtime polymorphism** using RBI as the parent class and SBI, ICICI, and PNB as child classes with different rates of interest.

System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

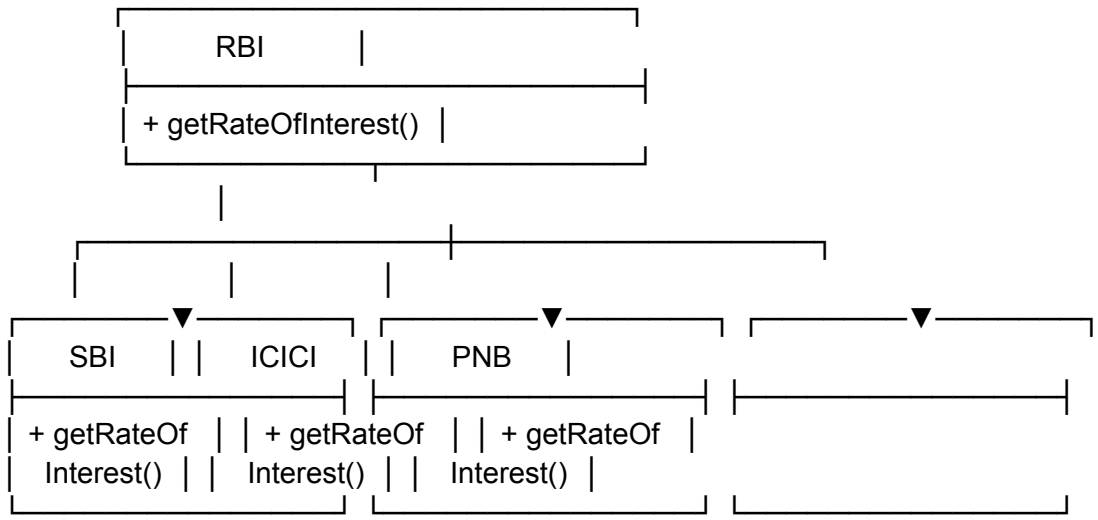
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or later)
- Text Editor or IDE such as VS Code, Eclipse, IntelliJ IDEA, or NetBeans
- Command Prompt / Terminal

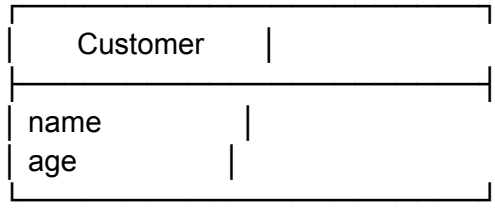
Algorithm / Logic Description

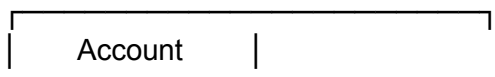
1. Create a `Customer` class with name and age attributes.
2. Create an `Account` class with account number and balance attributes.

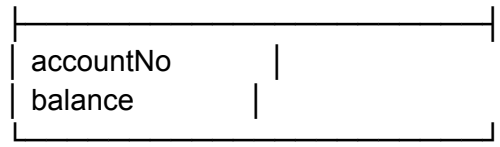
3. Create a parent class named `RBI`.
4. Define the `getRateOfInterest()` method in the `RBI` class.
5. Create child classes named `SBI`, `ICICI`, and `PNB` that inherit from `RBI`.
6. Override the `getRateOfInterest()` method in each child class with its respective interest rate.
7. Create a reference variable of the parent class `RBI`.
8. Read the bank name from the user.
9. According to the entered bank name, create the corresponding child class object.
10. Store the child class object in the `RBI` reference variable.
11. Call `getRateOfInterest()` using the `RBI` reference.
12. Due to method overriding, the corresponding child class method is executed at runtime.
13. Display the rate of interest.

UML Class Diagram









Relationship: `SBI`, `ICICI`, and `PNB` inherit from the `RBI` class and override the `getRateOfInterest()` method.

Source Code

```
import java.util.Scanner;

// creating a class for customer
class Customer {

    String name;
    int age;

    // constructor
    Customer(String name, int age) {
        this.name = name;
        this.age = age;
    }
}


// creating a class for account
class Account {

    int accountNo;
    double balance;

    // constructor
    Account(int accountNo, double balance) {
        this.accountNo = accountNo;
        this.balance = balance;
    }
}


// creating parent class RBI
```

```
class RBI {

    // method to return rate of interest
    double getRateOfInterest() {
        return 4.0;
    }
}


// creating SBI class
class SBI extends RBI {

    // overriding getRateOfInterest method
    double getRateOfInterest() {
        return 7.0;
    }
}


// creating ICICI class
class ICICI extends RBI {

    // overriding getRateOfInterest method
    double getRateOfInterest() {
        return 6.5;
    }
}


// creating PNB class
class PNB extends RBI {

    // overriding getRateOfInterest method
    double getRateOfInterest() {
        return 6.0;
    }
}


// main class
public class Bank {

    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        // taking bank name from user
        System.out.print(
            "Enter the Bank name to find the rate of Interest : "
        );

        String name = sc.nextLine();

        // creating parent class reference
        RBI bank;

        // creating SBI object
        if (name.equalsIgnoreCase("SBI")) {
            bank = new SBI();
        }

        // creating ICICI object
        else if (name.equalsIgnoreCase("ICICI")) {
            bank = new ICICI();
        }

        // creating PNB object
        else if (name.equalsIgnoreCase("PNB")) {
            bank = new PNB();
        }

        // creating RBI object for other bank names
        else {
            bank = new RBI();
        }

        // calling overridden method
        System.out.println(
            name.toUpperCase() +
            " rate of interest is : " +
            bank.getRateOfInterest() + "%"
        );

        sc.close();
    }
}
```

Compilation & Execution Commands

Save the source code exactly as:

Bank.java

Compile the program:

javac Bank.java

Run the program:

java Bank

Output Screen

Example 1: SBI

Enter the Bank name to find the rate of Interest : SBI
SBI rate of interest is : 7.0%

Example 2: ICICI

Enter the Bank name to find the rate of Interest : ICICI
ICICI rate of interest is : 6.5%

Example 3: PNB

Enter the Bank name to find the rate of Interest : PNB
PNB rate of interest is : 6.0%

Example 4: Other Bank

Enter the Bank name to find the rate of Interest : ABC
ABC rate of interest is : 4.0%

```
PS C:\Users\navee\OneDrive\Desktop\programs> javac Week7.java
PS C:\Users\navee\OneDrive\Desktop\programs> java Week7
Enter the Bank name to find the rate of Interest : sbi
SBI rate of interest is : 7.0%
PS C:\Users\navee\OneDrive\Desktop\programs> java Week7
Enter the Bank name to find the rate of Interest : cgi
CGI rate of interest is : 4.0%
PS C:\Users\navee\OneDrive\Desktop\programs> java Week7
Enter the Bank name to find the rate of Interest : ghj
GHJ rate of interest is : 4.0%
```